



JKBOSE 2026

CLASS 11TH

Relations and Functions

MATHS

LECTURE-05

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TOPICS TO BE COVERED

1 Some Functions and their Graphs

2 Algebra of functions

3 ~~Algebra of functions~~

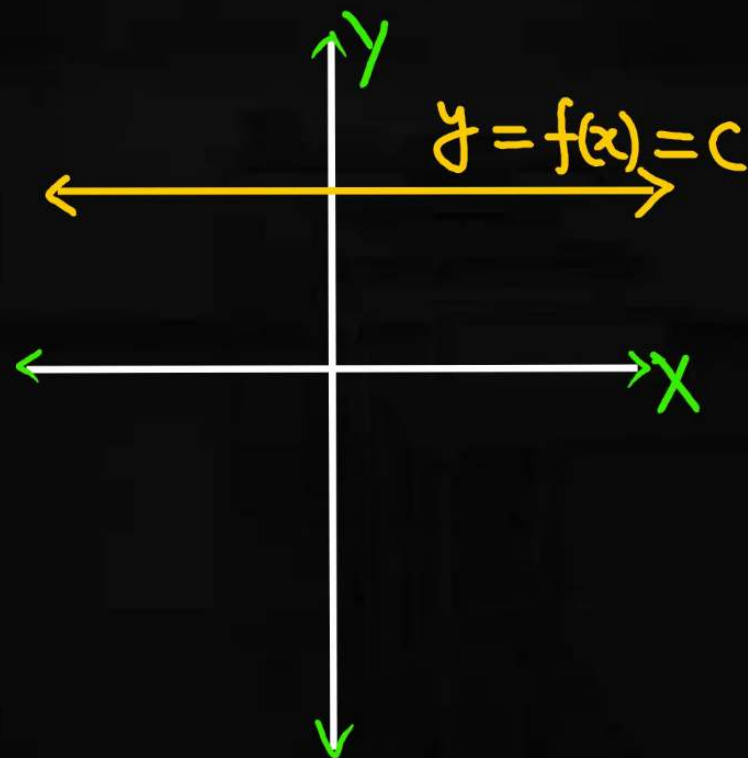
3 Related Problems

Some functions and their graphs

* **Constant function**: A function f defined by $f(x) = c$ for all $x \in \mathbb{R}$, where c is a fixed Real number, is known as a constant function.

* Domain = $\mathbb{R}$

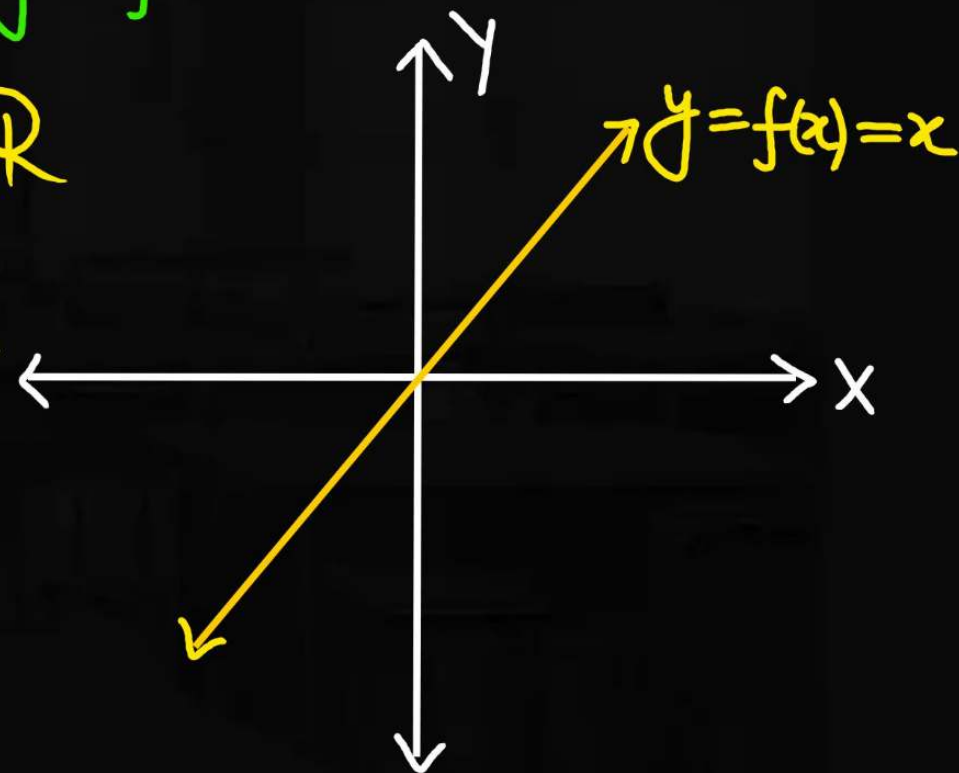
* Range = $\{c\}$



* **Identity function**: A function f defined by $f(x) = x$ for all $x \in \mathbb{R}$ is known as an Identity function.

* Domain = $\mathbb{R}$

* Range = $\mathbb{R}$



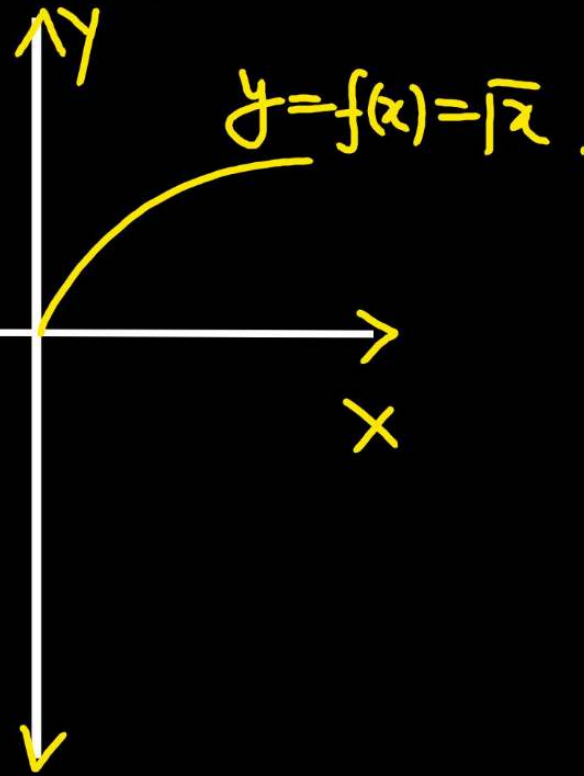
* **Square root function**: A function f defined by

$$f(x) = \sqrt{x}$$

is known as square root function.

* Domain = $[0, \infty)$

* Range = $[0, \infty)$



* **Polynomial function**: A function f

$$\text{defined by } f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

where $a_n (\neq 0), a_{n-1}, \dots, a_1, a_0 \in \mathbb{R}$

and n is non-negative integer is known as a polynomial function of Degree n .

* $f(x) = 2x^2 + x + 1$ (Quadratic polynomial)

* Graph of the Quadratic polynomial is a Parabola.

* $f(x) = x^2 + x^{1/2} + 2$ (not a polynomial)

Rational function: A function f of the form $* f(x) = \frac{2x^2+x}{x^3+x}$ is a Rational function.

$$f(x) = \frac{P(x)}{Q(x)}, \text{ where } P(x) \text{ and } Q(x) \neq 0$$

are Polynomials, is known as a Rational function.

$$* \text{ Domain} = \{x \in \mathbb{R} : Q(x) \neq 0\}$$

$$\Rightarrow \text{Domain} = \mathbb{R} - \{x \in \mathbb{R} : Q(x) = 0\}$$

Modulus function: A function f defined

$$\text{by } f(x) = |x| = \begin{cases} x & ; x \geq 0 \\ -x & ; x < 0 \end{cases} \text{ is known}$$

as the modulus function (Absolute value function)

$$* f(1) = |1| = 1$$

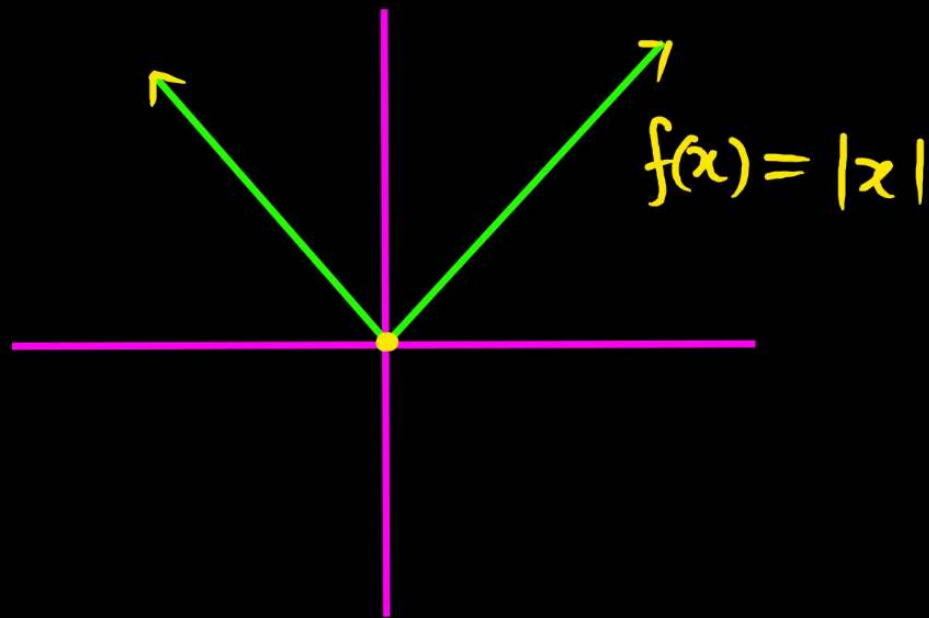
$$f(-1) = |-1| = -(-1) = 1$$

$$f(2) = |2| = 2$$

$$f(-2) = |-2| = -(-2) = 2$$

$$* f\left(-\frac{1}{2}\right) = \left|-\frac{1}{2}\right| = \frac{1}{2}$$

$$* f(0) = |0| = 0$$



* Domain = $\mathbb{R}$

* Range = $[0, \infty)$

$$\text{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

* **Signum function**: A function f defined by

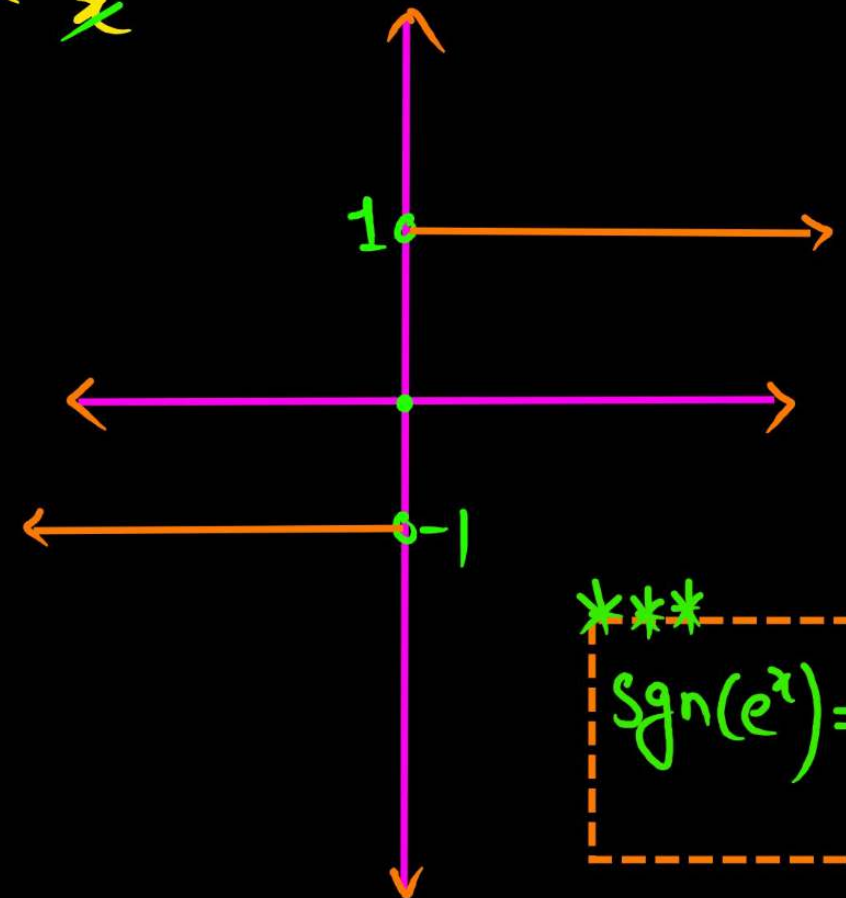
$$f(x) = \begin{cases} \frac{|x|}{x} & ; x \neq 0 \\ 0 & ; x = 0 \end{cases} = \begin{cases} \frac{x}{x} & ; x > 0 \\ 0 & ; x = 0 \\ -\frac{x}{x} & ; x < 0 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} 1 & ; x > 0 \\ 0 & ; x = 0 \\ -1 & ; x < 0 \end{cases}$$

* Also denoted by $\text{sgn}(x)$.

* Domain = $\mathbb{R}$

* Range = $\{-1, 0, 1\}$



$$\text{sgn}(e^x) = 1$$

Greatest integer function: A function f defined

by $f(x) = [x]$, where $[x]$ denotes the

greatest integer less or equal to x .

$$* f(1.2) = [1.2] = 1$$

$$* f(10.2) = [10.2] = 10$$

$$* f(5) = [5] = 5$$

$$* f(11.9) = [11.9] = 11$$

$$* f(-1.2) = [-1.2] = -2.$$

$$* f(-2.9) = [-2.9] = -3.$$

$$* \text{for } -3 \leq x < -2; [x] = -3$$

$$* \text{for } -2 \leq x < -1; [x] = -2$$

$$* \text{for } -1 \leq x < 0; [x] = -1$$

$$* \text{for } 0 \leq x < 1, [x] = 0$$

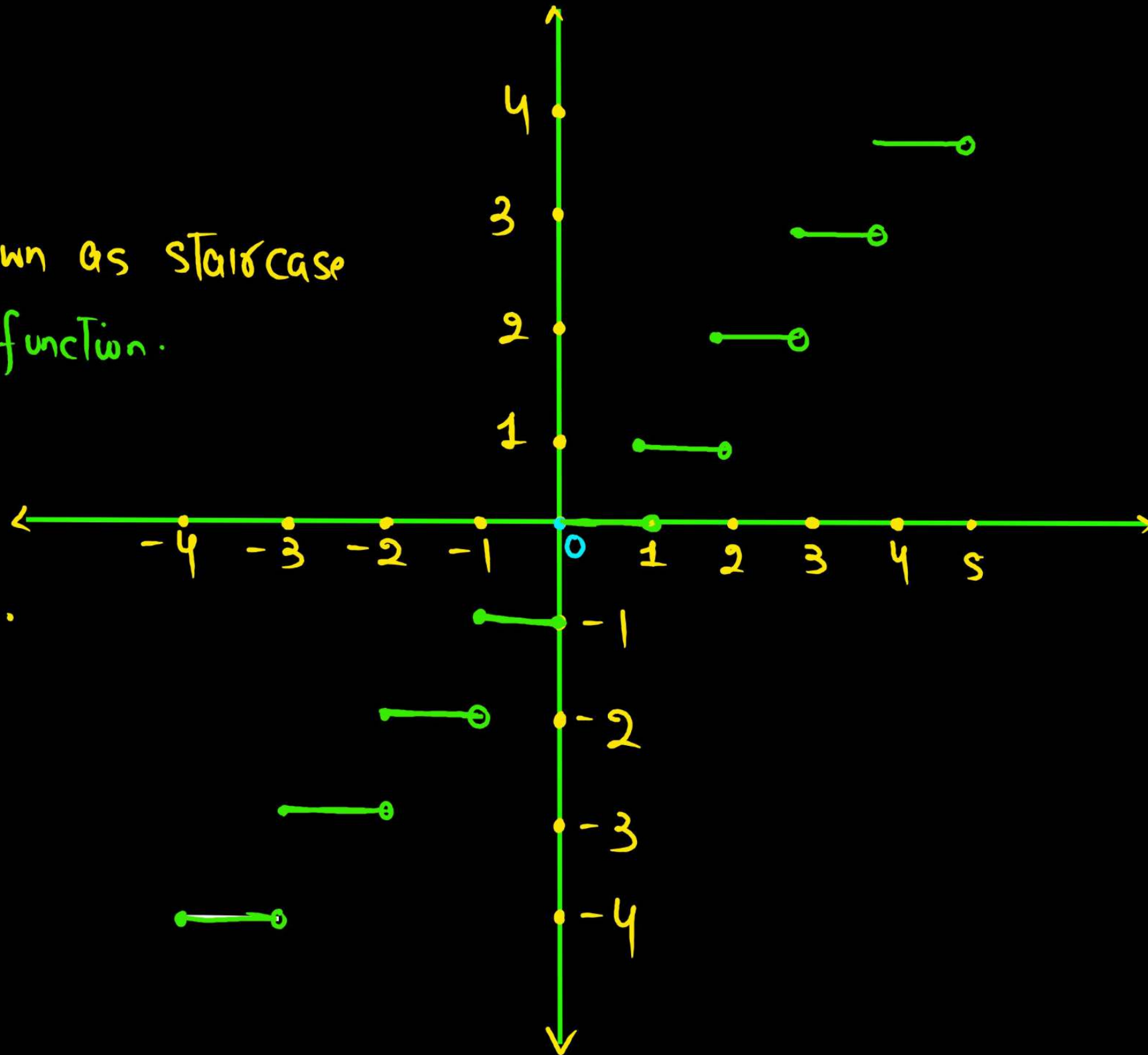
$$* \text{for } 1 \leq x < 2; [x] = 1$$

$$* \text{for } 2 \leq x < 3; [x] = 2.$$

* It is also known as staircase function / Floor function.

* Domain = $\mathbb{R}$.

* Range = $\mathbb{Z}$.



Algebra of Functions

$$*(f + g)(x) = f(x) + g(x)$$

$$*(f - g)(x) = f(x) - g(x)$$

$$*(fg)(x) = f(x)g(x)$$

$$*\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, g(x) \neq 0$$

Example

Let $f(x) = x^2$ and $g(x) = 2x + 1$ be two real functions.

Find $(f + g)(x)$, $(f - g)(x)$, $(fg)(x)$, $\left(\frac{f}{g}\right)(x)$

$$(Sol) (f + g)(x) = f(x) + g(x) = x^2 + 2x + 1$$

$$(f - g)(x) = f(x) - g(x) = x^2 - 2x - 1$$

$$(fg)(x) = f(x)g(x) = x^2(2x + 1) = 2x^3 + x^2$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{x^2}{2x + 1}, \quad x \neq -\frac{1}{2}$$

$$[2.3] = 2$$

$$2 \leq x < 3$$

$$[x] = 2.$$

$$2x^2 + x + 1$$

$$2x^2 + x + 1 = 0$$

Thank You

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class 11th - 12th

functions -> Real valued
functions